AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 5 · WORKBOOK

Inspect Before Modeling

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One coherent inspection-and-provenance record.

Explain counts, source limits, flag overlap and evidence-supported changes.

Keep one habit: W1–W3/W5 → fresh export → project inspection record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Python and Table Readiness

> **My remembered idea, example and correction**

Features, Labels, Targets

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 12

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Robot practice v1 is teacher-authored fictional indoor trip data, intended one row per trip, durations in minutes and None unknown. Rows (A,8),(B,None),(B,None),(C,-2),(D,12) give 5 raw rows,4 IDs,2 missing,1 negative,1 extra ID occurrence; two rows belong to repeated-ID group B and both are missing. Subtracting all flags double-counts.

  Preserve the raw version and unresolved source questions; no split or model score exists. Replacing None with0 or -2 with2 invents evidence. Copying rows adds neither new trip conditions nor independent observations.

- For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Robot practice v1 is teacher-authored fictional indoor trip data, intended one row per trip, durations in minutes and None unknown. Rows (A,8),(B,None),(B,None),(C,-2),(D,12) give 5 raw rows,4 IDs,2 missing,1 negative,1 extra ID occurrence; two rows belong to repeated-ID group B and both are missing. Subtracting all flags double-counts.

  Preserve the raw version and unresolved source questions; no split or model score exists. Replacing None with0 or -2 with2 invents evidence. Copying rows adds neither new trip conditions nor independent observations.

- Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Robot practice v1 is teacher-authored fictional indoor trip data, intended one row per trip, durations in minutes and None unknown. Rows (A,8),(B,None),(B,None),(C,-2),(D,12) give 5 raw rows,4 IDs,2 missing,1 negative,1 extra ID occurrence; two rows belong to repeated-ID group B and both are missing. Subtracting all flags double-counts.

  Preserve the raw version and unresolved source questions; no split or model score exists. Replacing None with0 or -2 with2 invents evidence. Copying rows adds neither new trip conditions nor independent observations.

- A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Robot practice v1 is teacher-authored fictional indoor trip data, intended one row per trip, durations in minutes and None unknown. Rows (A,8),(B,None),(B,None),(C,-2),(D,12) give 5 raw rows,4 IDs,2 missing,1 negative,1 extra ID occurrence; two rows belong to repeated-ID group B and both are missing. Subtracting all flags double-counts.

  Preserve the raw version and unresolved source questions; no split or model score exists. Replacing None with0 or -2 with2 invents evidence. Copying rows adds neither new trip conditions nor independent observations.

- Someone **duplicate**s each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected task uses its own six-row fictional indoor export, not the five-row main example or height picture. The schema intends one row per trip; duration is numeric nonnegative minutes or None for unknown. An extra ID occurrence means raw-row count minus distinct-ID count. List overlapping row flags before any unique-record conclusion; do not silently clean the supplied raw version.

- A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

  Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Inspect and document independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Interpret the audit counts for Dataset inspection and cards. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.
- Name checks and a deletion risk.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 5 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

- A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

  Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

▼ **Extension · Units conversions and data provenance — W1 Retain the source**

REFERENCE · SECTION 17

Extension · Units conversions and data provenance — W1 Retain the source

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Name four pieces of information to keep for Log A and its **conversion**.

My reasoning and evidence

▼ **Extension · Units conversions and data provenance — W2 Convert precisely**

REFERENCE · SECTION 18

Extension · Units conversions and data provenance — W2 Convert precisely

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

My reasoning and evidence

▼ **Extension · Units conversions and data provenance — W3 Repair a comparison**

REFERENCE · SECTION 19

Extension · Units conversions and data provenance — W3 Repair a comparison

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A report claims 120 seconds exceeds 3 minutes because 120 > 3. Correct it.

My reasoning and evidence

▼ Extension · Units conversions and data provenance — W4 Normalize independently

REFERENCE · SECTION 20

Extension · Units conversions and data provenance — W4 Normalize independently

> Retained extension task with its own values and teacher key.

Extension question and evidence

- Create records for 240 seconds, 1.5 minutes and 60 seconds. Derive minutes while keeping raw data. State the values and their total.

My reasoning and evidence

▼ Extension · Units conversions and data provenance — W5 Handle an unknown unit

REFERENCE · SECTION 21

Extension · Units conversions and data provenance — W5 Handle an unknown unit

> Retained extension task with its own values and teacher key.

Extension question and evidence

- A record contains value 8 and unit ticks with no **conversion** rule. Should you assume seconds?

> **My reasoning and evidence**

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.
- Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/5

Inspect Before Modeling

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count rows

Five raw rows represent four distinct IDs; B is duplicated.



2. Check schema

C violates the nonnegative-height rule; D=0 is valid under this contract.



3. Flag before fitting

Flag duplicate B and invalid C. Do not count B twice or silently replace C.

CHANGE ONE THING

The teacher confirms B's second row is an accidental duplicate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 24

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 25

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.
- Which labelled step matches this detail? Flag **duplicate** B and invalid C. Do not count B twice or silently replace C.
- 1. Flag before **fitting**
- 2. Count rows
- 3. Check **schema**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
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> ---
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> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Flag before **fitting** Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. Read the picture in order. Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The teacher confirms B's second row is an accidental **duplicate**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Repair and keep the record

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

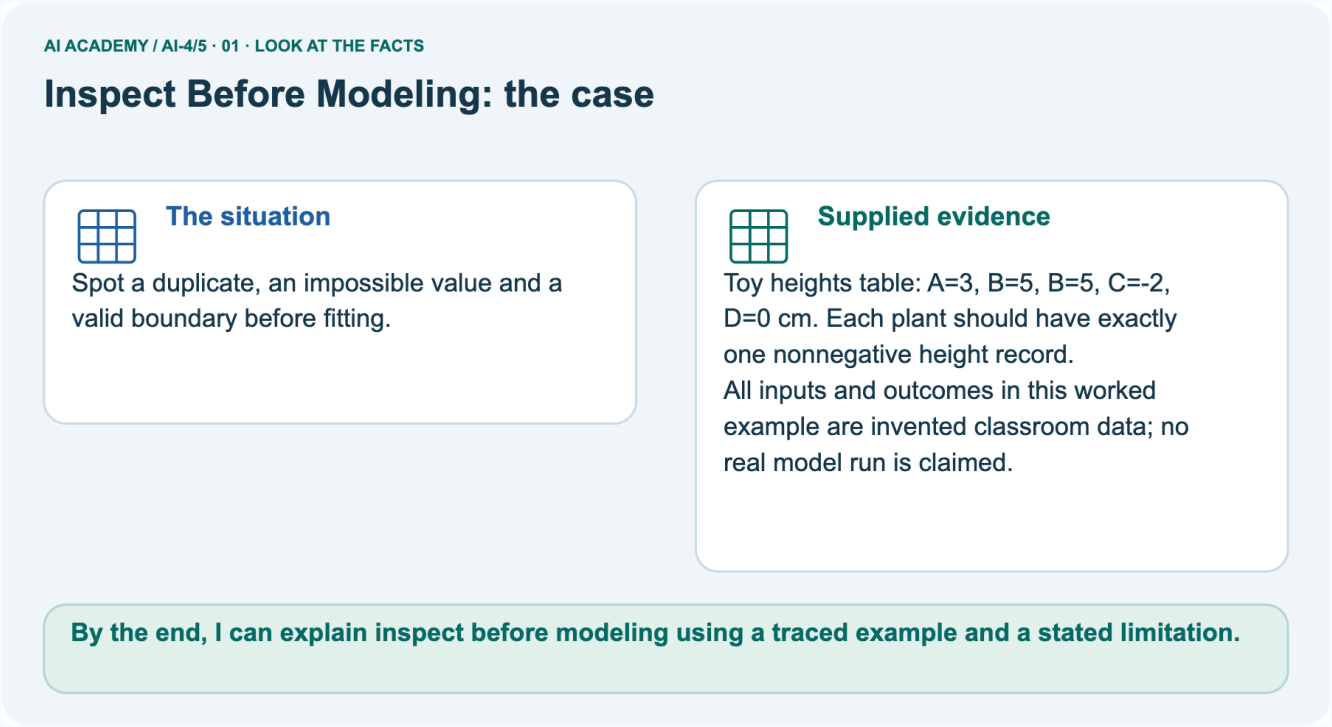
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Inspect Before Modeling: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

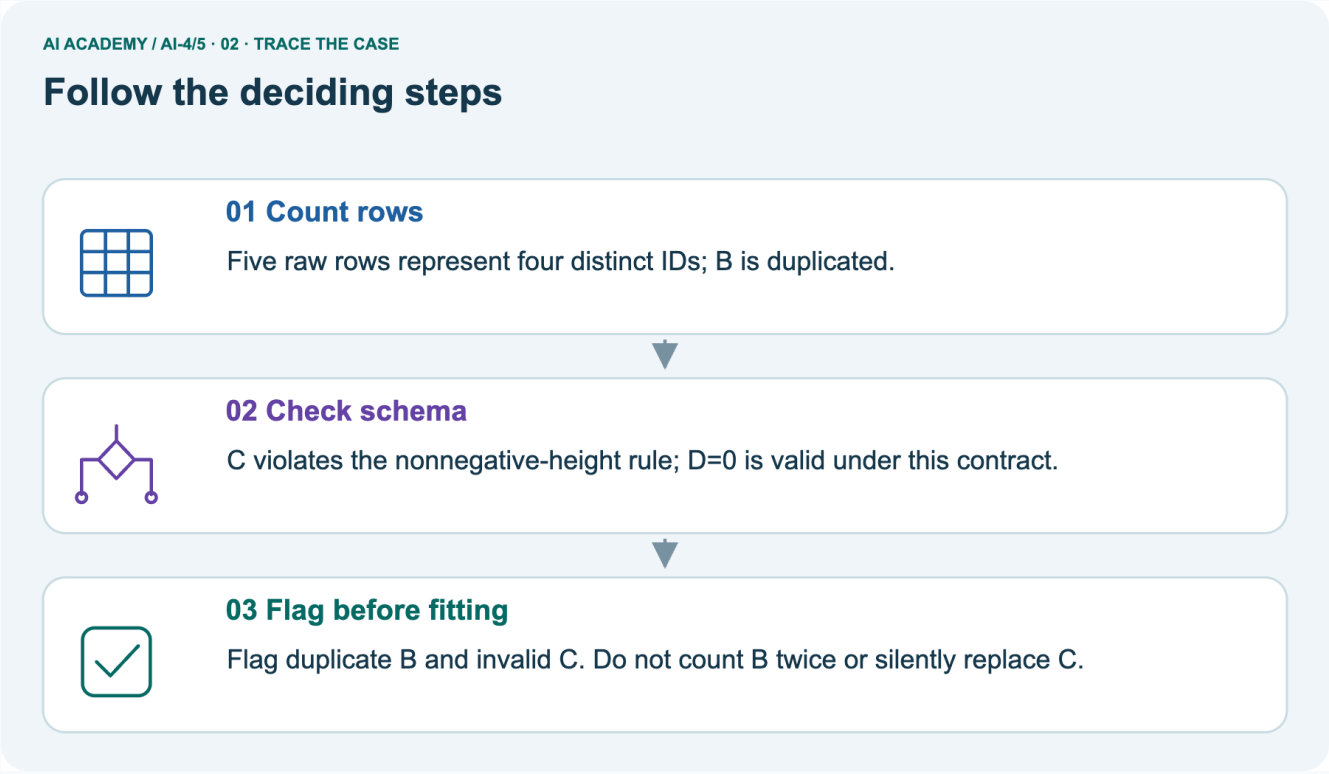
Spot a **duplicate**, an impossible value and a valid boundary before **fitting**. Supplied facts: Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Count rows: Five raw rows represent four distinct IDs; B is **duplicate**d. → Check **schema**: C violates the nonnegative-height rule; D=0 is valid under this contract. → Flag before **fitting**: Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action

AI ACADEMY / AI-4/5 · 03 · INSPECT THE LESSON MODEL

Turn an observation into a documented action

Finding	What it establishes	Next action
B appears twice	Identity needs review	Check export history
Two None values	2 of 5 rows lack duration	Keep unknown explicit
C has minus 2	Invalid elapsed duration	Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

Follow the labelled picture with the matching explanation. Open the vector diagram

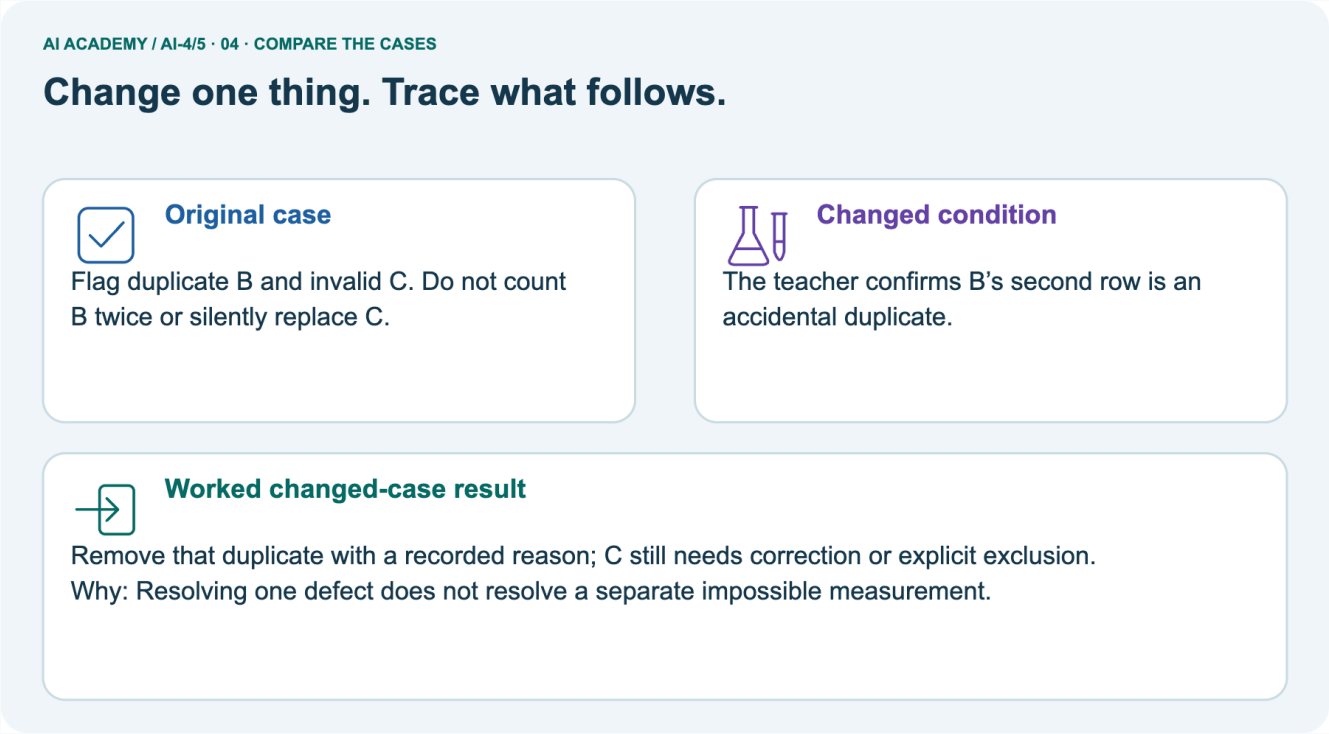
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

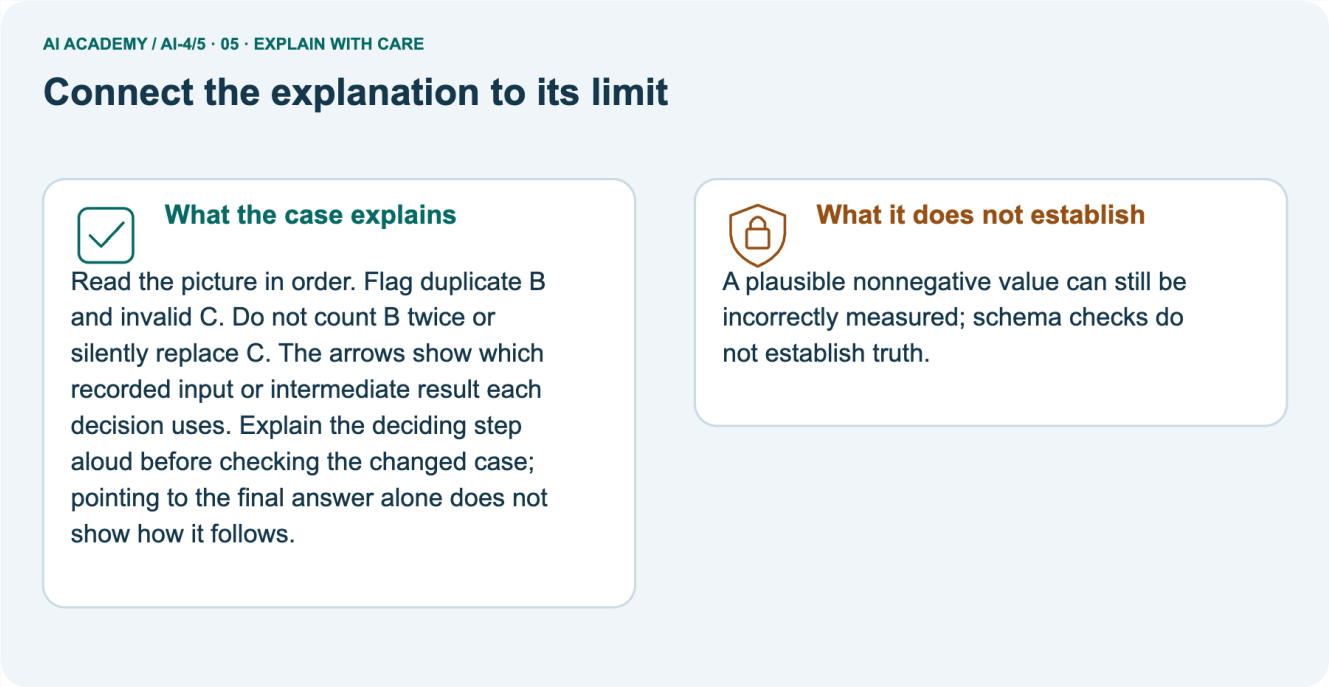
Original case: Flag **duplicate** B and invalid C. Do not count B twice or silently replace C.
Changed condition: The teacher confirms B's second row is an accidental **duplicate**.

Expected result: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

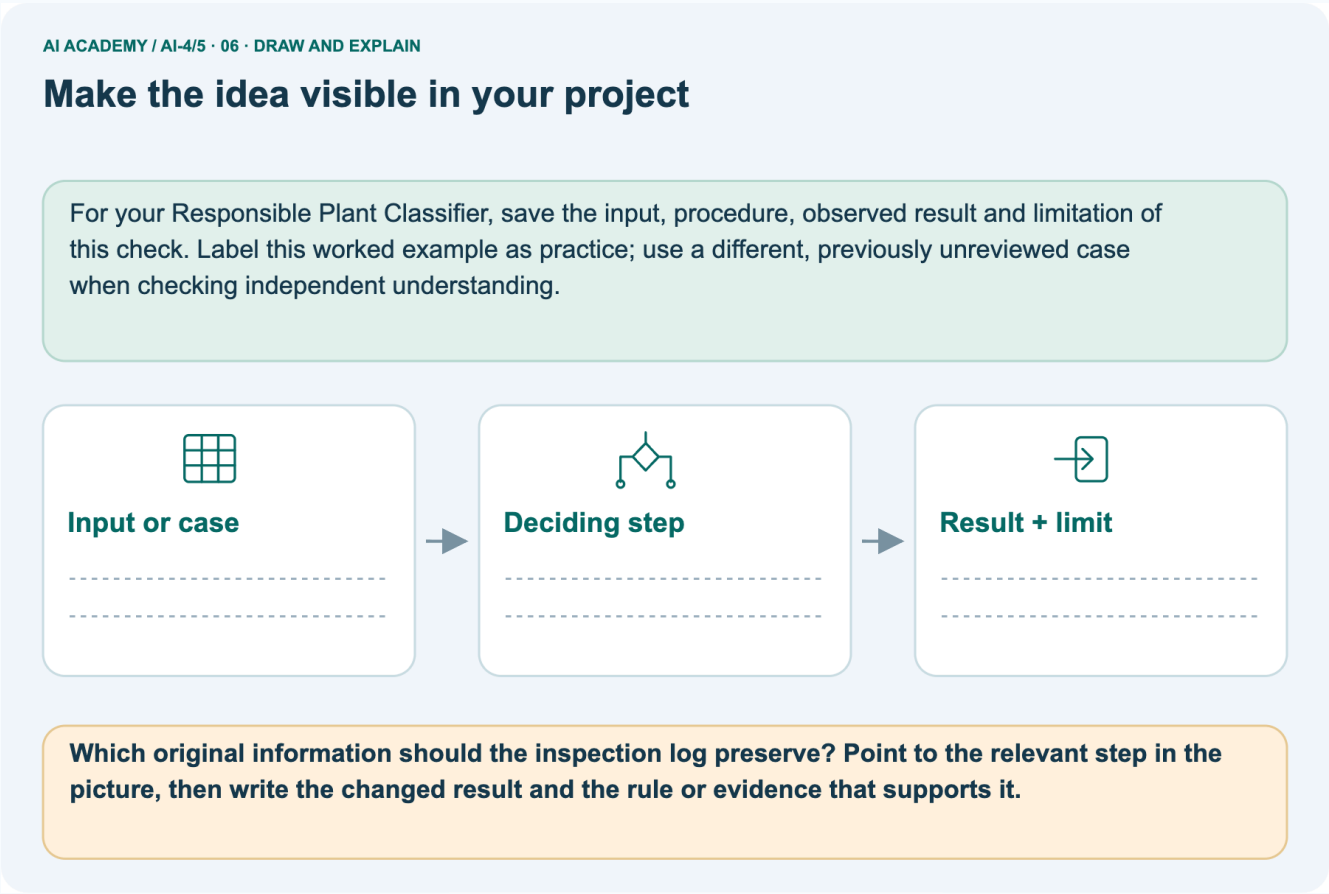
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Inspect Before Modeling: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.